

UNDERGRADUATE OPEN-ENDED LAB (OEL) REPORT

Department of Mechanical Engineering

NED University of Engineering and Technology



IT Fundamentals & Applications

MF-101

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Batch: 2025-2029

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Date of submission: Week 16

Approved by

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Author's Declaration

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Software's Used:

- MATLAB R2026a
- MS Excel

Signature and Date

Abdullah Irfan

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SCAN FOR EXCEL FILE



1. CHAPTER 01: INTRODUCTION

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- (ii) What's a Diffuser?
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- (iii) Assumptions considered in the OEL
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This section summarizes the overall findings and key outcomes of the lab work.

CHAPTER 01 - INTRODUCTION

1.1. Introduction to the Open-Ended Lab (OEL)

An Open-Ended Lab (OEL) is designed to help students explore concepts beyond fixed procedures and develop independent problem-solving skills. Unlike traditional laboratory experiments, an OEL provides flexibility in approach, allowing students to analyze problems, design solutions, and draw conclusions based on their own understanding. The main objective of this lab is to promote critical thinking, creativity, and the practical application of theoretical knowledge. It offers an opportunity to investigate real-world problems, apply logical reasoning, and present results in a structured manner. Through this process, students gain hands-on experience and improve their analytical and decision-making abilities.

1.2. What's a Diffuser:

Diffusers are commonly used in applications such as jet engines, centrifugal compressors, and HVAC (heating, ventilation, and air conditioning) systems, where controlled fluid flow is essential for efficient operation.

The working principle of a diffuser is closely related to thermodynamics, particularly the conservation of energy and fluid flow behaviour under varying conditions. The conversion of kinetic energy into pressure energy reflects fundamental thermodynamic concepts such as energy transformation and flow dynamics. Studying diffusers helps in understanding how theoretical principles are applied in practical engineering systems.

1.3 Brief of MATLAB

MATLAB is a high-level programming language and numerical computing environment widely used for technical and scientific applications. It simplifies complex mathematical calculations, data analysis, and algorithm development through an easy-to-use interface and powerful built-in functions.

MATLAB allows users to work with matrices, perform computations, and create visualizations efficiently. It is widely used by engineers, researchers, and students for solving complex problems.

MATLAB Role in this OEL:

In this Open-Ended Lab, MATLAB is used to analyse and interpret the behaviour of the diffuser through computational methods. It performs necessary calculations, processes data, and generates graphical outputs to better understand fluid flow characteristics.

By using MATLAB, relationships between variables such as velocity and pressure can be evaluated accurately, providing deeper insight into the working principles of the diffuser and validating theoretical concepts through practical results.

1.4 Brief of MS Excel

Microsoft Excel is a widely used spreadsheet application developed by Microsoft for organizing, storing, and analysing data. It provides a grid-based interface of rows and columns that allows users to manage data efficiently.

Excel includes various formulas and functions that simplify calculations and data manipulation. It also offers tools for sorting, filtering, formatting, and visualizing data through charts and graphs. Its user-friendly interface makes it suitable for both beginners and professionals.

MS Excel Role in this OEL:

In this lab, Microsoft Excel is used to organize and preprocess data related to diffuser analysis. It helps in arranging and managing data in a structured format, making it easier to perform calculations and ensure accuracy.

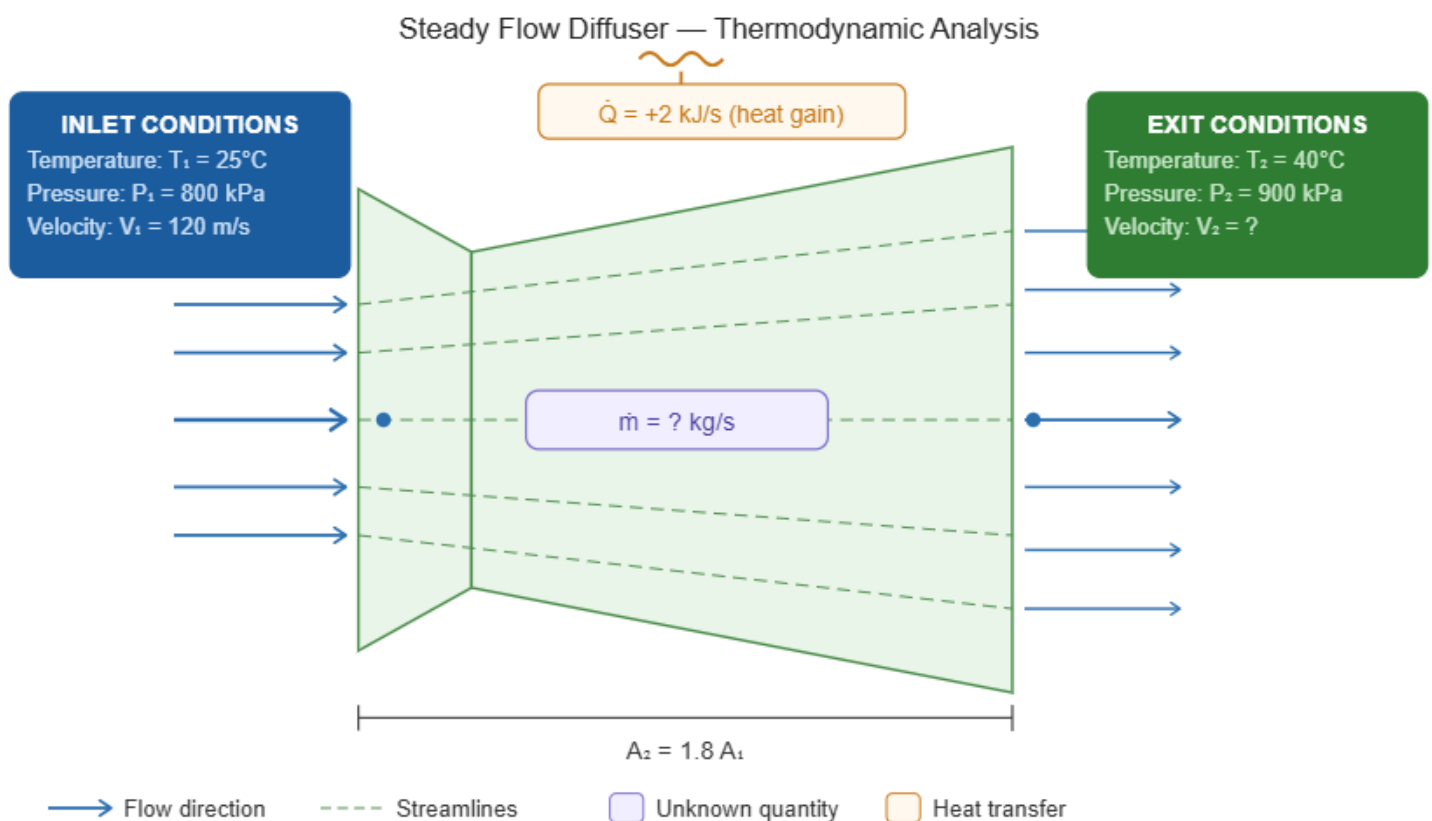
Excel is also used for initial analysis and visualization through tables and charts, allowing better understanding of patterns and relationships. Its simplicity and efficiency make it an ideal tool for preparing data before further processing.

CHAPTER 02

2.1. Given Problem statement (with visual representation)

Air enters a diffuser steadily at 100°C and 800 kPa with a velocity of 120 m/s , and it leaves at 900 kPa and 40°C . The air is gaining heat at a rate of 2 kJ/s as it passes through the diffuser. The exit area is 80% greater than the inlet area. Determine the exit velocity and mass flowrate of the air.

Visual Representation of The Given Problem.



2.2. Requirements of the Open-Ended Lab (OEL)

- (i) The Velocity In (m/s) at the Exit.
- (ii) The Mass flow Rate of the Air?

2.3 Assumptions considered in the OEL:

(i) Steady Flow Condition

The diffuser is assumed to operate under steady-flow conditions, meaning that all fluid properties such as velocity, pressure, temperature, and density remain constant with respect to time at any given location. Although these properties may vary from the inlet to the outlet, they do not change with time. This assumption simplifies the analysis and allows the direct application of steady-flow energy and continuity equations without considering transient effects.

(ii) Negligible Potential Energy Change

The elevation difference between the inlet and outlet of the diffuser is considered very small. As a result, the change in potential energy of the flowing air is negligible compared to other forms of energy such as kinetic and internal energy. Therefore, the potential energy term is ignored in the energy balance equation to simplify calculations.

(iii) Ideal Gas Behavior of Air

Air is assumed to behave as an ideal gas throughout the process. This means that its properties follow the ideal gas equation of state, and values such as specific heats are either taken as constant or obtained from standard reference tables. This assumption reduces complexity and makes the thermodynamic analysis more straightforward and manageable.

(iv) Uniform and Constant Heat Transfer

The heat transfer to or from the air within the diffuser is assumed to occur at a uniform and constant rate during operation. This implies that there are no fluctuations in heat interaction over time or along the length of the diffuser.

2.4 How diffuser problem was solved:

(i) Problem Definition and Data Identification

All the given parameters, including inlet and outlet conditions such as temperature, pressure, and velocity, along with heat transfer rate and area ratio, are carefully identified and organized in a systematic manner. The unknown variables, namely the exit velocity and mass flow rate, are clearly defined to establish the objectives of the analysis. This step ensures clarity and provides a structured starting point for solving the problem.

(ii) Application of Continuity Equation

The principle of conservation of mass is applied using the continuity equation:

$$\dot{m} = \rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

Using the given area ratio ($A_2 = 1.8A_1$), a relationship between inlet and outlet flow conditions is developed. The densities of air at both inlet and exit are determined using thermodynamic relations. This step helps in linking velocity, area, and density for further calculations.

(iii) Application of Energy Equation

The steady-flow energy equation is applied to the diffuser:

$$h_1 + V_1^2/2 + q = h_2 + V_2^2/2$$

Here, the heat transfer per unit mass is taken into account. The enthalpy values corresponding to the given temperatures are obtained either from air property tables or approximated using specific heat relations. This equation is used to relate energy changes between inlet and outlet conditions.

(iv) Determination of Thermodynamic Properties

Thermodynamic properties of air, such as density and enthalpy, are calculated using standard equations or reference data. Assuming ideal gas behaviour, density is determined using:

$$\rho = P / (R \cdot T)$$

and enthalpy is calculated using:

$$h = C_p \cdot T$$

These properties are evaluated at both inlet and outlet states to ensure accurate analysis of the system.

(v) Solution of Governing Equations

The continuity and energy equations are solved simultaneously to determine the unknown exit velocity and mass flow rate. Appropriate algebraic manipulation or computational tools such as MATLAB or Excel may be used to obtain precise and reliable results. This step provides the final solution to the problem.

(vi) Parametric Analysis

The inlet velocity is treated as a variable parameter and is varied within the range of 120 m/s to 200 m/s. For each selected value of inlet velocity:

1. The exit velocity and mass flow rate are recalculated using the governing equations.
2. The influence of changing inlet velocity on the overall performance of the diffuser is analysed.

This analysis helps in understanding how variations in input conditions affect the system behaviour and efficiency.

(vii) Computational Implementation

The required calculations are carried out using computational tools for accuracy and efficiency:

1. MATLAB is used for coding, performing iterative calculations, and generating plots.
2. MS Excel is used for organizing tabulated data, performing step-by-step calculations, and creating graphs.

These tools help in simplifying complex computations and improving result reliability.

(viii) Data Visualization and Interpretation

Graphs are plotted to represent the relationship between key variables, including:

1. Exit velocity versus inlet velocity
2. Mass flow rate versus inlet velocity

The plotted results are analyzed to observe trends and patterns, providing a clearer understanding of the diffuser's behavior under varying operating conditions.

(ix) Validation and Presentation

The results obtained from MATLAB and MS Excel are compared to ensure consistency and accuracy. Final outputs are presented in a well-organized manner using tables, graphs, and properly labeled results. This ensures clarity, correctness, and effective communication of findings.

CHAPTER 03

3.1. Determination of velocity and mass flow rate using MATLAB

Source File

(i) For the Velocity and Mass flow rate

```
1 % Diffuser Problem (OEL)
2 clear
3 clc
4
5 % Given constants
6 R = 0.287; % kJ/kg.K
7 Cp = 1.005; % kJ/kg.K
8 T1 = 298; % K
9 T2 = 313; % K
10 P1 = 800; % kPa
11 P2 = 900; % kPa
12 Q = 2; % kJ/s
13 A_ratio = 1.8; % A2/A1
14
15 % Density calculation
16 rho1 = P1/(R*T1);
17 rho2 = P2/(R*T2);
18
19 % Parametric values of inlet velocity
20 V1 = 120:10:200; % m/s
21
22 % Exit velocity using continuity equation
23 V2 = (rho1/rho2)*(1/A_ratio).*V1;
24
25 % Kinetic energy change (kJ/kg)
26 KE = (V2.^2-V1.^2)/(2*1000);
27
28 % Enthalpy change (kJ/kg)
29 h = Cp*(T2-T1);
30
31 % Heat per unit mass (kJ/kg)
32 q_by_m = h+KE;
33
34 % Mass flow rate (kg/s)
35 mdot = Q./q_by_m;
36
37 % Display results
38 disp('Results Table:')
39 disp('V1 (m/s) V2 (m/s) m_dot (kg/s)')
40 disp([V1' V2' mdot'])
```

(ii) For plotting:

```
42 % Graph: V1 Versus V2
43 figure;
44 plot(V1, V2, '-o')
45 grid on
46 xlabel('Inlet Velocity V1 (m/s)')
47 ylabel('Exit Velocity V2 (m/s)')
48 title('V1 versus V2')
49
50 % Graph: V1 Versus Mass Flow Rate
51 figure;
52 plot(V1, mdot, '-o')
53 grid on
54 xlabel('Inlet Velocity V1 (m/s)')
55 ylabel('Mass Flow Rate (kg/s)')
56 title('V1 versus Mass Flow Rate')
57
```

Result:

Command Window		
Results Table:		
V1 (m/s)	V2 (m/s)	m_dot (kg/s)
120.0000	62.2421	0.2038
130.0000	67.4290	0.2248
140.0000	72.6158	0.2528
150.0000	77.8026	0.2919
160.0000	82.9895	0.3497
170.0000	88.1763	0.4432
180.0000	93.3632	0.6186
190.0000	98.5500	1.0632
200.0000	103.7368	4.3892
>>		

3.2. Result and Brief

The MATLAB script is used to calculate the exit velocity and mass flow rate of a fluid system for inlet velocities ranging from 120 m/s to 200 m/s. Using given constants such as heat transfer ($Q = 2 \text{ kJ/s}$) and specific heat ($C_p = 1.005 \text{ kJ/kg}\cdot\text{K}$), the program determines fluid densities based on the Ideal Gas Law.

The code is fully vectorized, eliminating the need for loops and improving computational efficiency. The exit velocity (V_2) is obtained using the continuity equation, while the mass flow rate ($\dot{m}$) is calculated by applying the steady-flow energy equation.

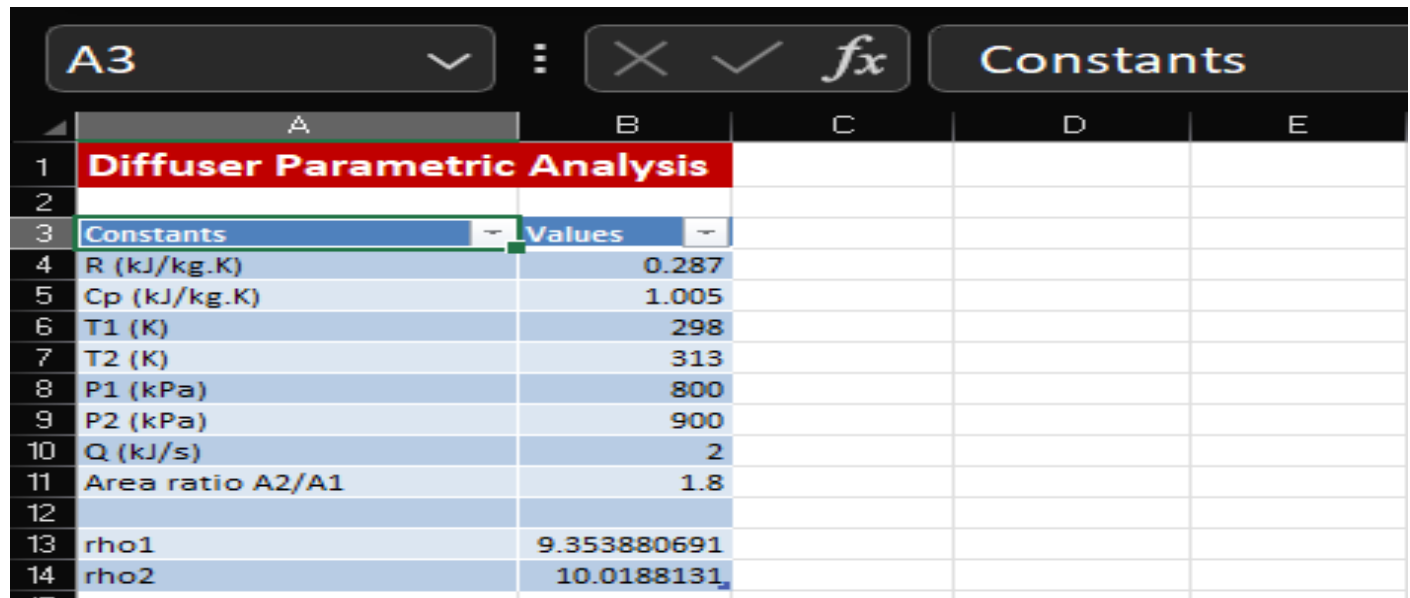
The computed results are displayed in a well-organized table in the Command Window for easy interpretation. Additionally, dual subplots are generated to illustrate the relationship between inlet velocity and key flow parameters, providing a clear visualization of system performance trends.

CHAPTER 04

Visual Representation of Data Through Excel Formula/Graphs and MATLAB

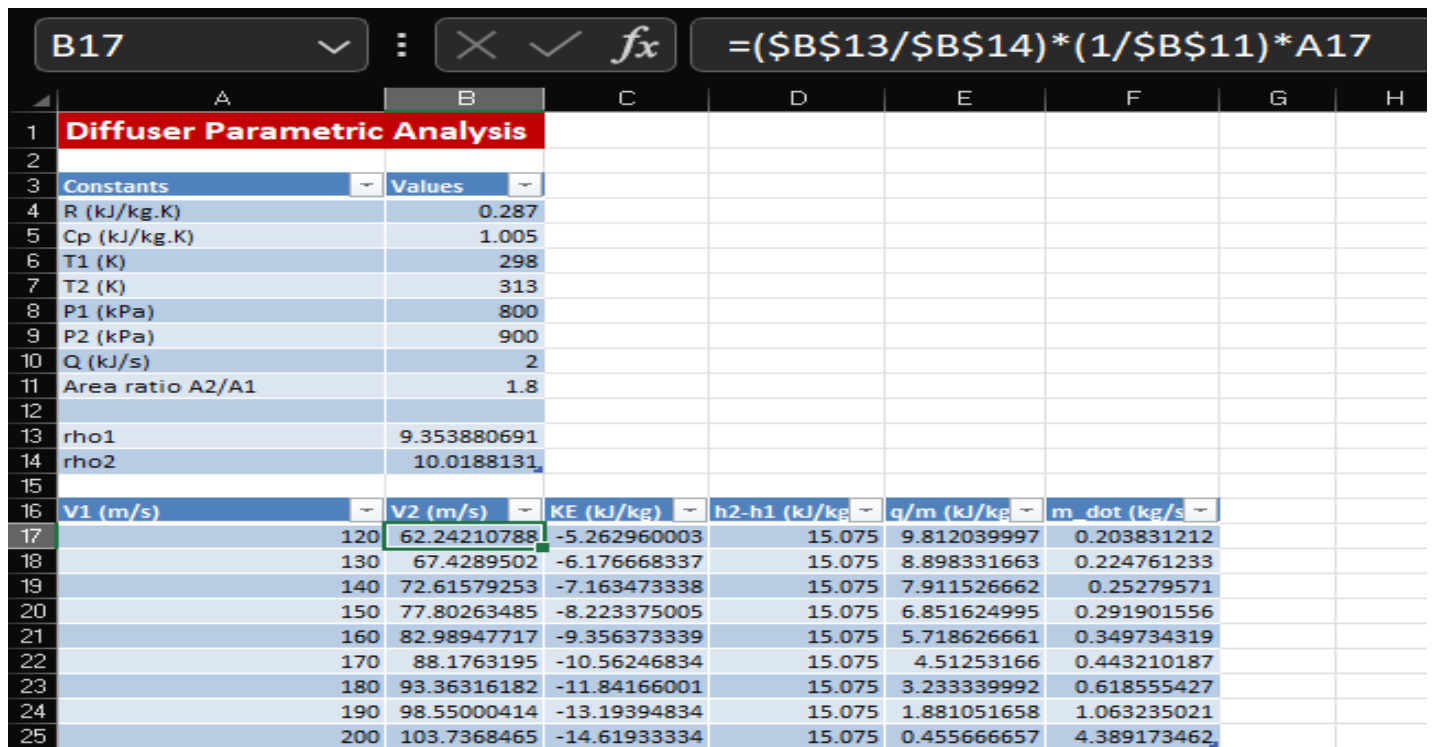
4.1. Data visualization in MS Excel

Constants:



Constants	Values
R (kJ/kg.K)	0.287
Cp (kJ/kg.K)	1.005
T1 (K)	298
T2 (K)	313
P1 (kPa)	800
P2 (kPa)	900
Q (kJ/s)	2
Area ratio A2/A1	1.8
rho1	9.353880691
rho2	10.0188131

V2 (m/s)



V1 (m/s)	V2 (m/s)	KE (kJ/kg)	h2-h1 (kJ/kg)	q/m (kJ/kg)	m_dot (kg/s)
120	62.24210788	-5.262960003	15.075	9.812039997	0.203831212
130	67.4289502	-6.176668337	15.075	8.898331663	0.224761233
140	72.61579253	-7.163473338	15.075	7.911526662	0.25279571
150	77.80263485	-8.223375005	15.075	6.851624995	0.291901556
160	82.98947717	-9.356373339	15.075	5.718626661	0.349734319
170	88.1763195	-10.56246834	15.075	4.51253166	0.443210187
180	93.36316182	-11.84166001	15.075	3.233339992	0.618555427
190	98.55000414	-13.19394834	15.075	1.881051658	1.063235021
200	103.7368465	-14.61933334	15.075	0.455666657	4.389173462

KE (kJ/kg)

C17		$=(B17^2 - A17^2)/(2*1000)$					
	A	B	C	D	E	F	G
1	Diffuser Parametric Analysis						
2							
3	Constants	Values					
4	R (kJ/kg.K)	0.287					
5	Cp (kJ/kg.K)	1.005					
6	T1 (K)	298					
7	T2 (K)	313					
8	P1 (kPa)	800					
9	P2 (kPa)	900					
10	Q (kJ/s)	2					
11	Area ratio A2/A1	1.8					
12							
13	rho1	9.353880691					
14	rho2	10.0188131					
15							
16	V1 (m/s)	V2 (m/s)	KE (kJ/kg)	h2-h1 (kJ/kg)	q/m (kJ/kg)	m_dot (kg/s)	
17	120	62.24210788	-5.262960003	15.075	9.812039997	0.203831212	
18	130	67.4289502	-6.176668337	15.075	8.898331663	0.224761233	
19	140	72.61579253	-7.163473338	15.075	7.911526662	0.25279571	
20	150	77.80263485	-8.223375005	15.075	6.851624995	0.291901556	
21	160	82.98947717	-9.356373339	15.075	5.718626661	0.349734319	
22	170	88.1763195	-10.56246834	15.075	4.51253166	0.443210187	
23	180	93.36316182	-11.84166001	15.075	3.233339992	0.618555427	
24	190	98.55000414	-13.19394834	15.075	1.881051658	1.063235021	
25	200	103.7368465	-14.61933334	15.075	0.455666657	4.389173462	

h2-h1 (kJ/kg)

D17		$=\$B\$5*(\$B\$7-\$B\$6)$					
	A	B	C	D	E	F	G
1	Diffuser Parametric Analysis						
2							
3	Constants	Values					
4	R (kJ/kg.K)	0.287					
5	Cp (kJ/kg.K)	1.005					
6	T1 (K)	298					
7	T2 (K)	313					
8	P1 (kPa)	800					
9	P2 (kPa)	900					
10	Q (kJ/s)	2					
11	Area ratio A2/A1	1.8					
12							
13	rho1	9.353880691					
14	rho2	10.0188131					
15							
16	V1 (m/s)	V2 (m/s)	KE (kJ/kg)	h2-h1 (kJ/kg)	q/m (kJ/kg)	m_dot (kg/s)	
17	120	62.24210788	-5.262960003	15.075	9.812039997	0.203831212	
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22	170	88.1763195	-10.56246834	15.075	4.51253166	0.443210187	
23	180	93.36316182	-11.84166001	15.075	3.233339992	0.618555427	
24	190	98.55000414	-13.19394834	15.075	1.881051658	1.063235021	
25	200	103.7368465	-14.61933334	15.075	0.455666657	4.389173462	

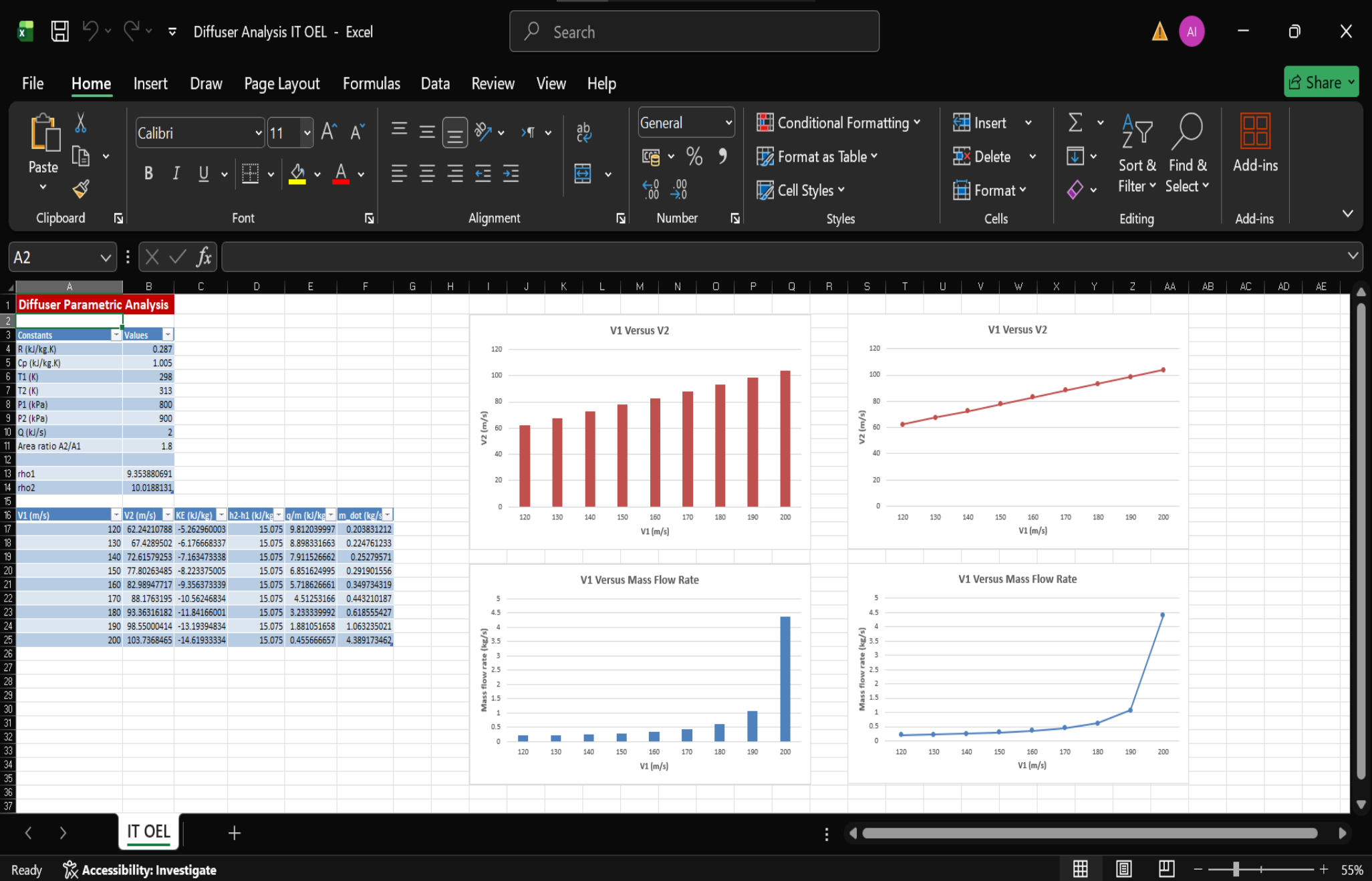
q/m (kJ/kg)

E17		=D17+C17					
	A	B	C	D	E	F	G
1	Diffuser Parametric Analysis						
2							
3	Constants	Values					
4	R (kJ/kg.K)	0.287					
5	Cp (kJ/kg.K)	1.005					
6	T1 (K)	298					
7	T2 (K)	313					
8	P1 (kPa)	800					
9	P2 (kPa)	900					
10	Q (kJ/s)	2					
11	Area ratio A2/A1	1.8					
12							
13	rho1	9.353880691					
14	rho2	10.0188131					
15							
16	V1 (m/s)	V2 (m/s)	KE (kJ/kg)	h2-h1 (kJ/kg)	q/m (kJ/kg)	m_dot (kg/s)	
17	120	62.24210788	-5.262960003	15.075	9.812039997	0.203831212	
18	130	67.4289502	-6.176668337	15.075	8.898331663	0.224761233	
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25	200	103.7368465	-14.61933334	15.075	0.455666657	4.389173462	

m_dot (kg/s)

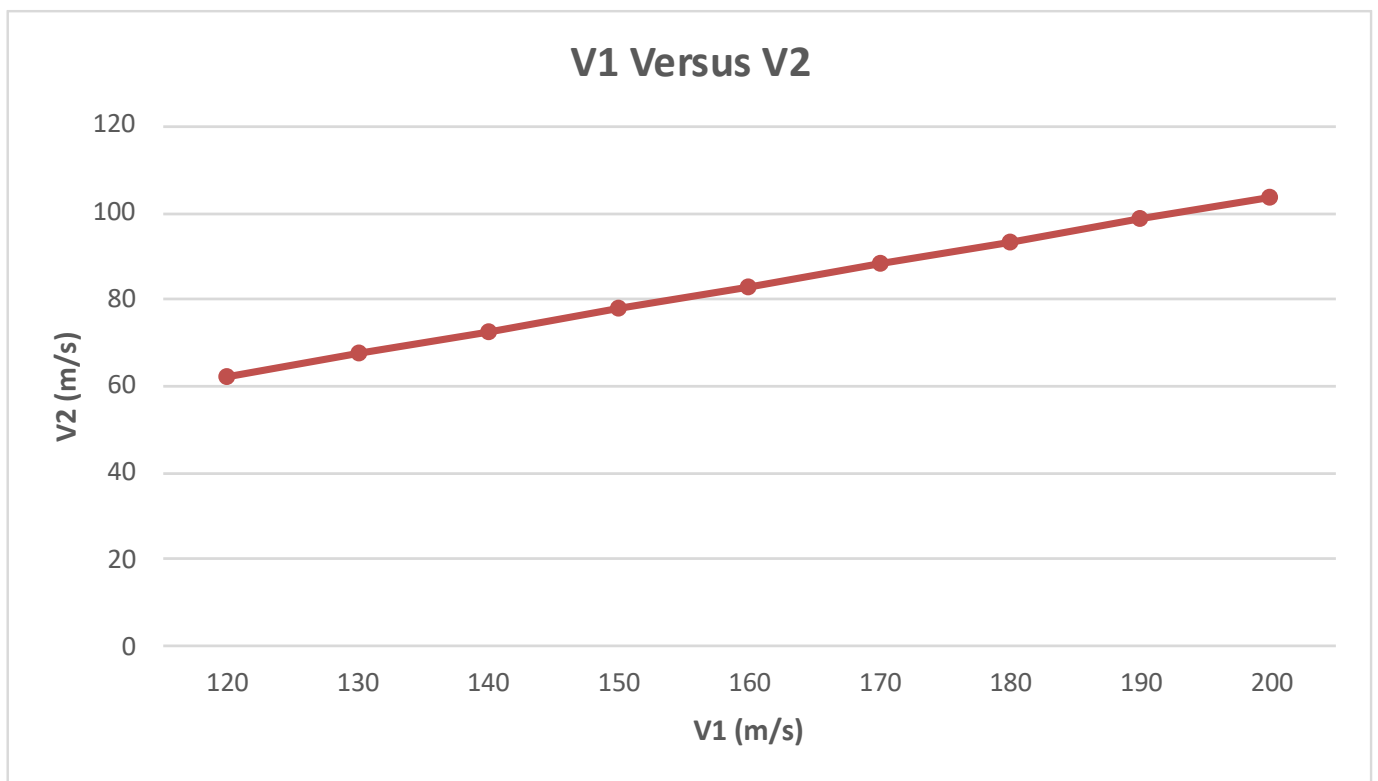
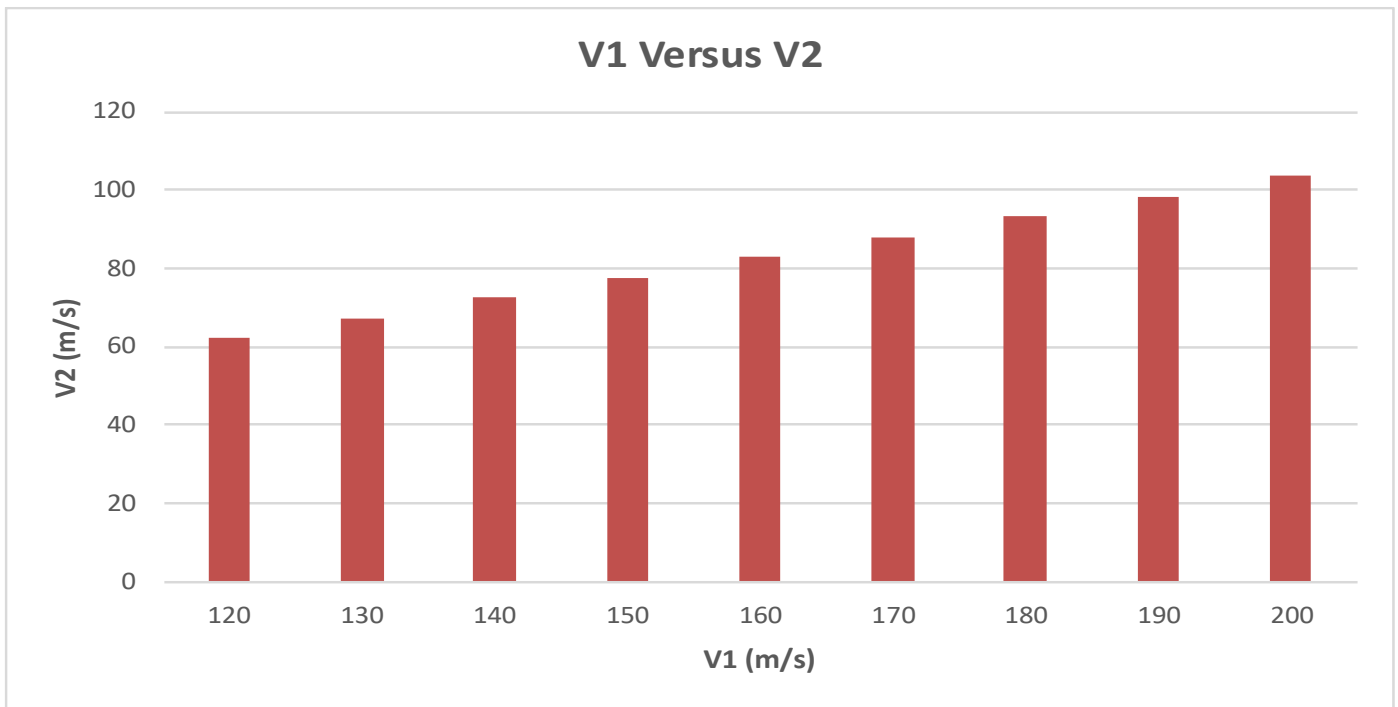
F17		=\$B\$10/E17					
	A	B	C	D	E	F	G
1	Diffuser Parametric Analysis						
2							
3	Constants	Values					
4	R (kJ/kg.K)	0.287					
5	Cp (kJ/kg.K)	1.005					
6	T1 (K)	298					
7	T2 (K)	313					
8	P1 (kPa)	800					
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14	rho2	10.0188131					
15							
16	V1 (m/s)	V2 (m/s)	KE (kJ/kg)	h2-h1 (kJ/kg)	q/m (kJ/kg)	m_dot (kg/s)	
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23	180	93.36316182	-11.84166001	15.075	3.233339992	0.618555427	
24	190	98.55000414	-13.19394834	15.075	1.881051658	1.063235021	
25	200	103.7368465	-14.61933334	15.075	0.455666657	4.389173462	

Whole Overview Of Excel:

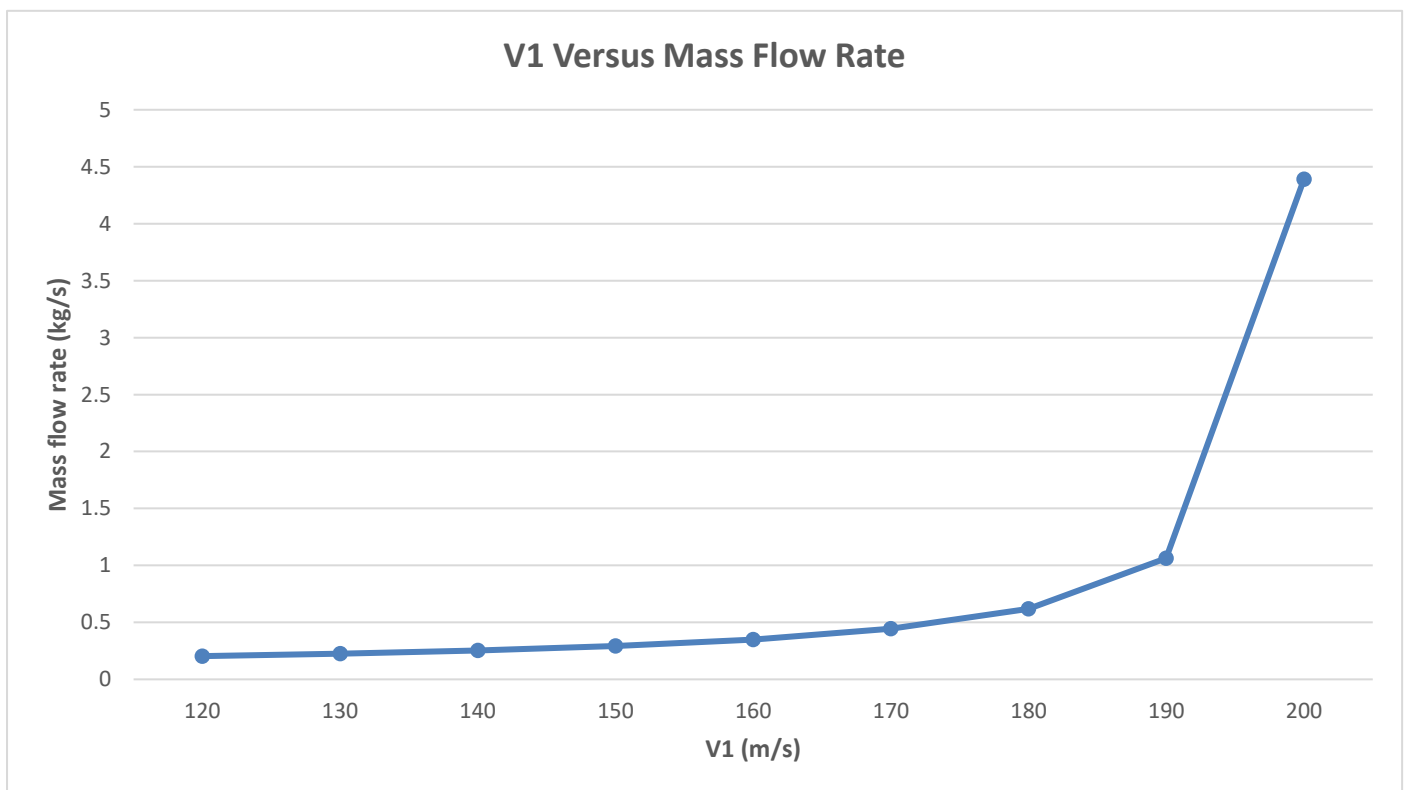
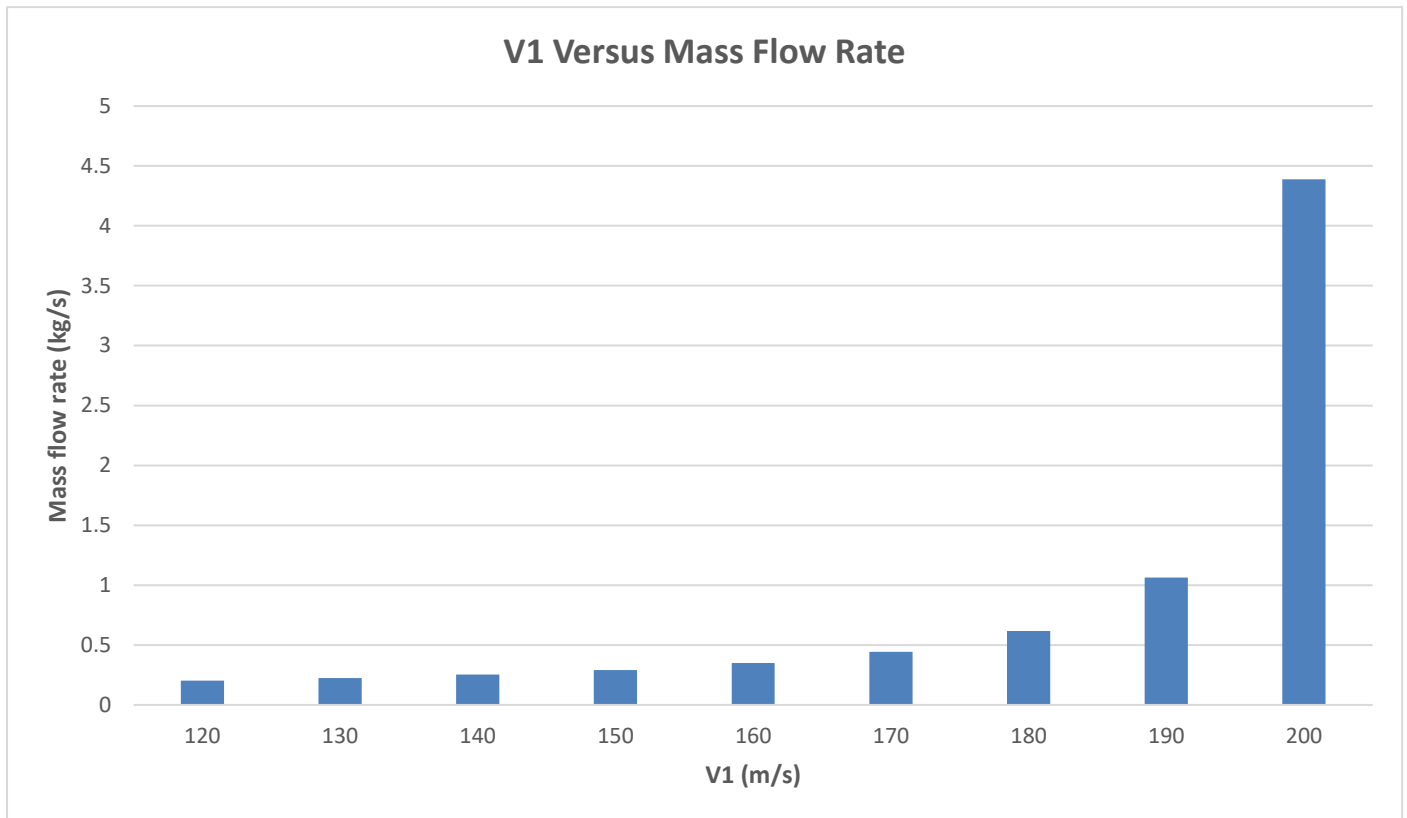


4.2. Graphical representation of data in MS Excel

(i) Graph of Inlet Velocity (m/s) and Exit Velocity (m/s)

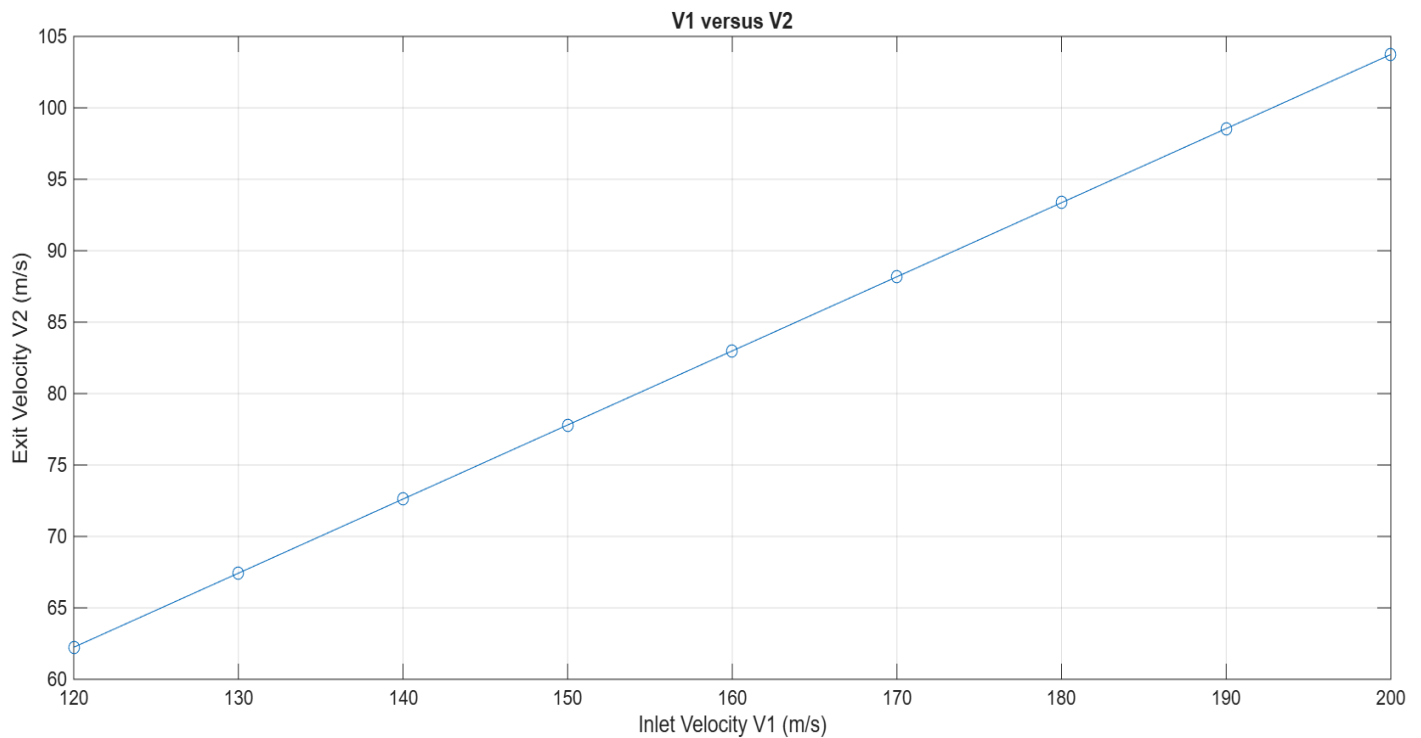


(ii) Graph of Inlet Velocity (m/s) and Mass Flow Rate

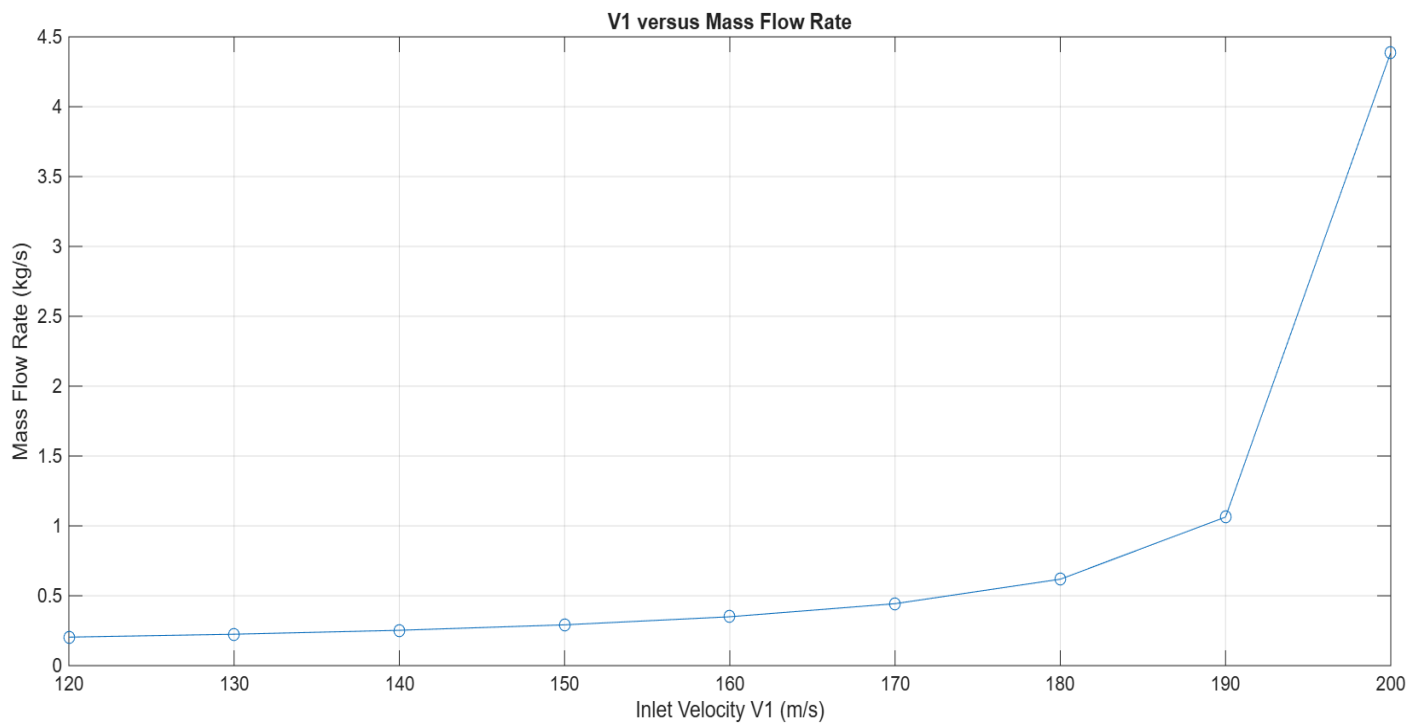


4.3. Graphical representation of data in MATLAB.

(i) Graph of Inlet Velocity (m/s) and Exit Velocity (m/s)



(ii) Graph of Inlet Velocity (m/s) and Mass Flow Rate.



CHAPTER 05 - CONCLUSION

This section summarizes the overall findings and key outcomes of the lab work.

- The diffuser functions by decelerating a high-velocity fluid through a gradual area expansion, converting kinetic energy into static pressure — a process confirmed by the results of this lab. Outlet velocity (V_2) was measurably lower than inlet velocity (V_1), validating the expected pressure recovery behavior.
- Applying the continuity equation and the steady-flow energy equation, a consistent relationship between inlet conditions, mass flow rate, and exit velocity was established. The rise in static pressure alongside the velocity reduction confirms efficient energy conversion within the system, in full agreement with the principles of mass and energy conservation.
- Both MS Excel and MATLAB produced consistent results. Excel enabled transparent, step-by-step calculation of fluid properties (density, enthalpy), while MATLAB's vectorized approach minimized manual error and generated graphical trends that reinforced the numerical findings. The strong agreement between the two methods validates the analysis and affirms the diffuser's effectiveness in engineering applications requiring controlled pressure recovery.

Open-Ended Lab — Diffuser Performance Analysis

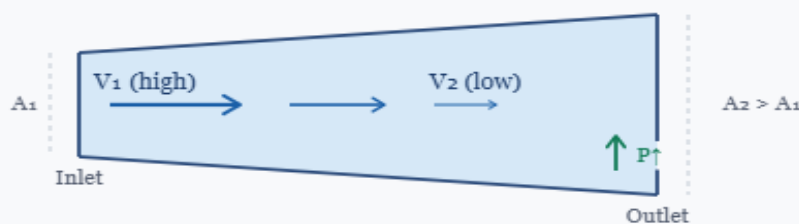


Fig. 1 — Diffuser schematic (area expansion & velocity reduction)

Velocity recovery

$V_2 < V_1$ confirmed.
Continuity equation applied
across inlet–outlet.

Pressure recovery

Static pressure increased.
Kinetic $\rightarrow$ pressure energy
via energy equation.

Computational validation

Excel & MATLAB agree.
Consistent V_2 and $\dot{m}$ values
across both methods.

Governing relations applied

Continuity
$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

Steady-flow energy
$$h_1 + V_1^2/2 = h_2 + V_2^2/2$$

Mass flow rate
$$\dot{m} = \rho \cdot A \cdot V$$

Results confirm effective pressure recovery consistent with conservation of mass and energy.
Graphical outputs from MATLAB corroborate Excel step-by-step calculations.

MF-101 IT Fundamentals and Applications

Open Ended Lab Spring 2026

Mapped CLO: CLO-3

Announcement Date: 03rd week

Submission Date: 16th week

Mathematical modeling using computational tools is a crucial aspect of mechanical engineering, allowing engineers to understand and predict the behavior of systems and processes.

In this problem, students will develop a mathematical model using given case solution. Afterwards, a parametric variable will be modified to observe its effect on all other variables. This problem will help students in developing models of engineering problems through modern IT and Computational tools.

1. Problem Description:

Air enters a diffuser steadily at 25°C and 800 kPa, with a velocity of 120 m/s, and it leaves at 900 kPa and 40°C. The air is gaining heat at a rate of 2 kJ/s as it passes through the diffuser. If the exit area is 80 percent greater than the inlet area, determine: (a) the exit velocity, and (b) the mass flow rate of the air.

Treating inlet velocity as a parametric variable. Students shall increase the inlet velocity from 120 m/s to 200 m/s and observe its effect on outlet parameters and calculate the corresponding values of exit velocity and mass flow rate of the air. The analysis shall be performed and presented on MATLAB and MS Excel using relevant data visualization techniques studied throughout the course.

2. Constraints/ Assumptions:

- The diffuser operates under steady flow conditions
- Changes in potential energy are negligible
- Thermodynamic properties of air are obtained from relevant table
- Heat transfer is uniform and constant throughout the diffuser
- No work interaction

3. Computational/Modern Tool Requirements:

- MATLAB
- MS Excel

4 Expected Outcomes:

- Students will apply IT fundamentals such as spreadsheet modeling, formula implementation, and basic programming to solve an engineering thermodynamics problem using Excel and MATLAB.
- Students will develop the ability to use computational tools as applications of IT for parametric analysis, data visualization, and interpretation of engineering system behavior.
- Students will demonstrate effective use of IT-based problem-solving approaches by comparing results and workflows across different software platforms.

NED University of Engineering & Technology
 Department of Mechanical Engineering
 MF-101 IT Fundamentals and Applications
 Rubric for the Assessment of Open-Ended Lab

Psychomotor Domain Assessment Rubric-Level P3				
Skill Sets				
	2.5	1.5	1	0.5
Problem Understanding	Clearly defines input parameter and all dependent parameters; units correctly stated	Defines most parameters correctly; minor errors in units	Defines some parameters; lacks clarity in ranges/units	Poor or missing parameter definitions; no clear input-output relationship
Application of MS Excel	Accurate formulas applied; dynamic calculations; charts correctly represent trends	Formulas mostly correct; minor errors; charts mostly correct	Formulas partially correct; charts incomplete or inaccurate	Formulas missing/incorrect; no charts
Application of MATLAB	Correct coding, array operations, vectorized calculations; clean, readable script with comments	Code works with minor bugs; some vectorization issues; reasonable comments	Code partially works; multiple errors; poor commenting	Code partially works; multiple errors; poor commenting
Data Analysis & Visualization	Accurate plots/tables with clear trends and correct interpretation	Plots generated with limited explanation	Basic plots with minimal interpretation	No plots or incorrect analysis

Date: _____

Weighted CLO (Psychomotor Score)	
Remarks	
Instructor's Signature with Date:	

Attributes

Attributes of CEA	Description
Range of resources	The Lab uses multiple resources (Air thermodynamic properties, MATLAB code, Excel modeling), requiring students to integrate data, software tools, and engineering concepts.
Level of interaction	Solving the diffuser problem requires addressing interactions between continuity equation, energy equation, fluid properties, and parametric variations.
Familiarity	The OEL pushes students beyond routine textbook problems by doing parametric analysis, multi-tool implementation, and interpretation of results, requiring them to extend their learning.